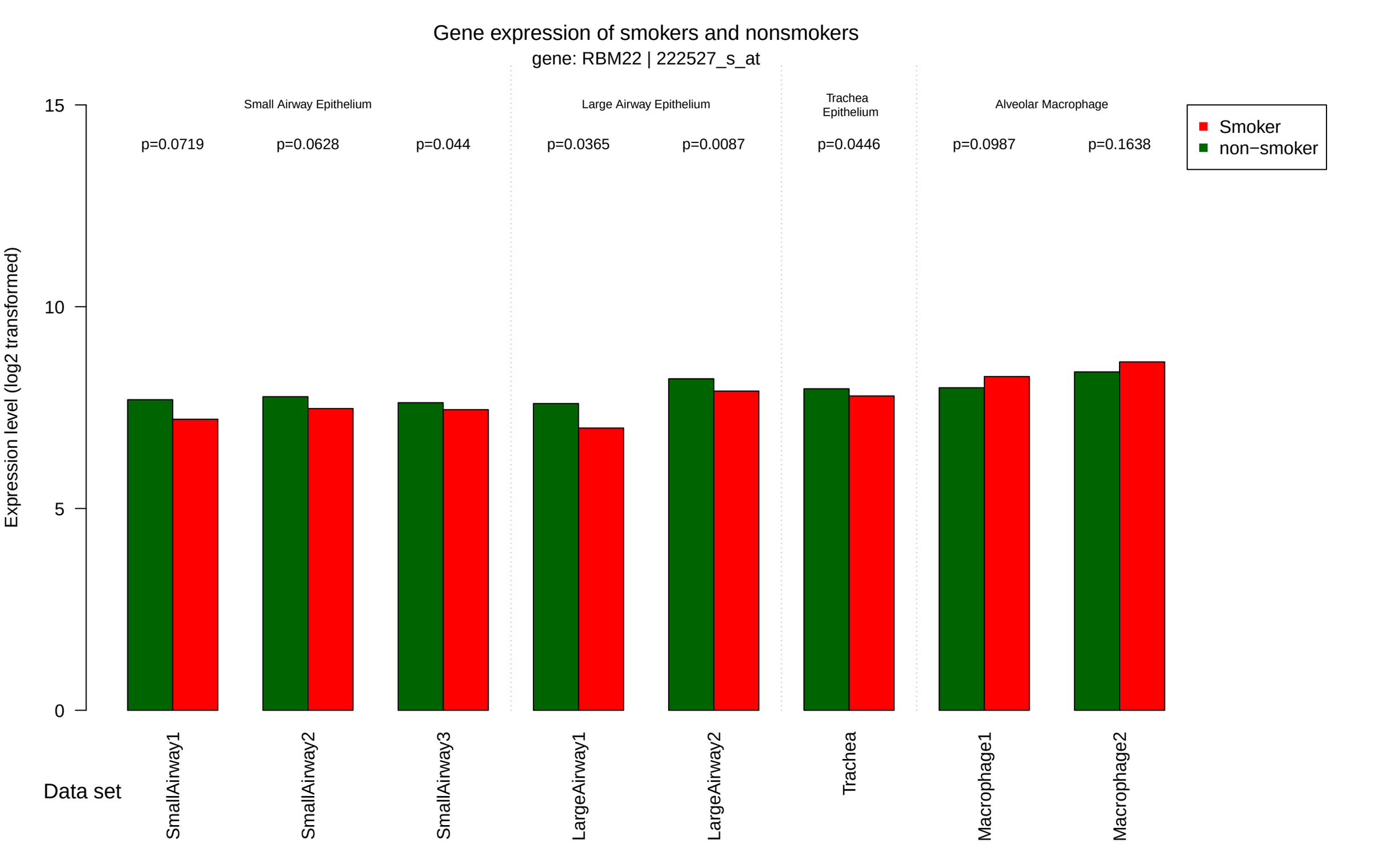
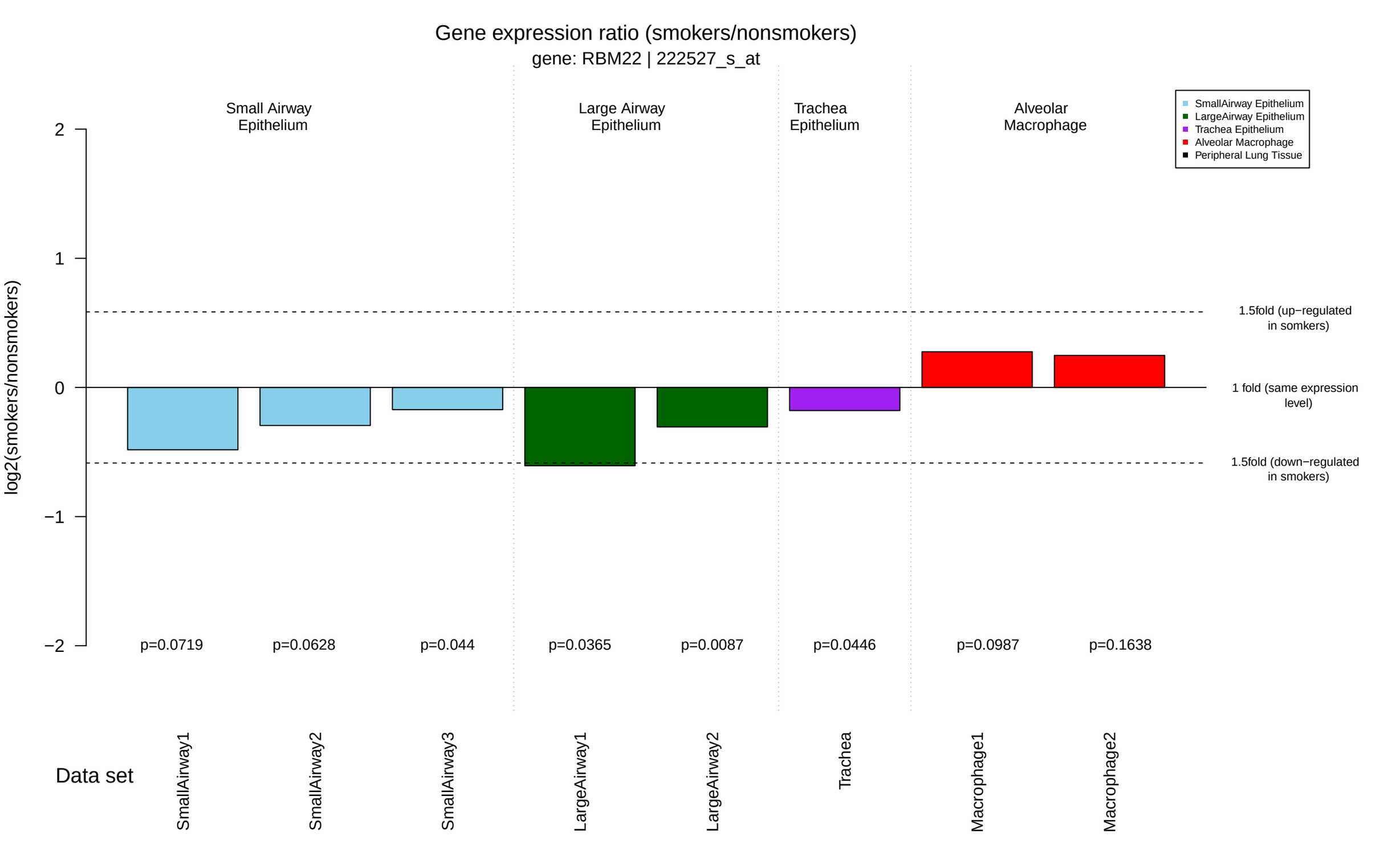
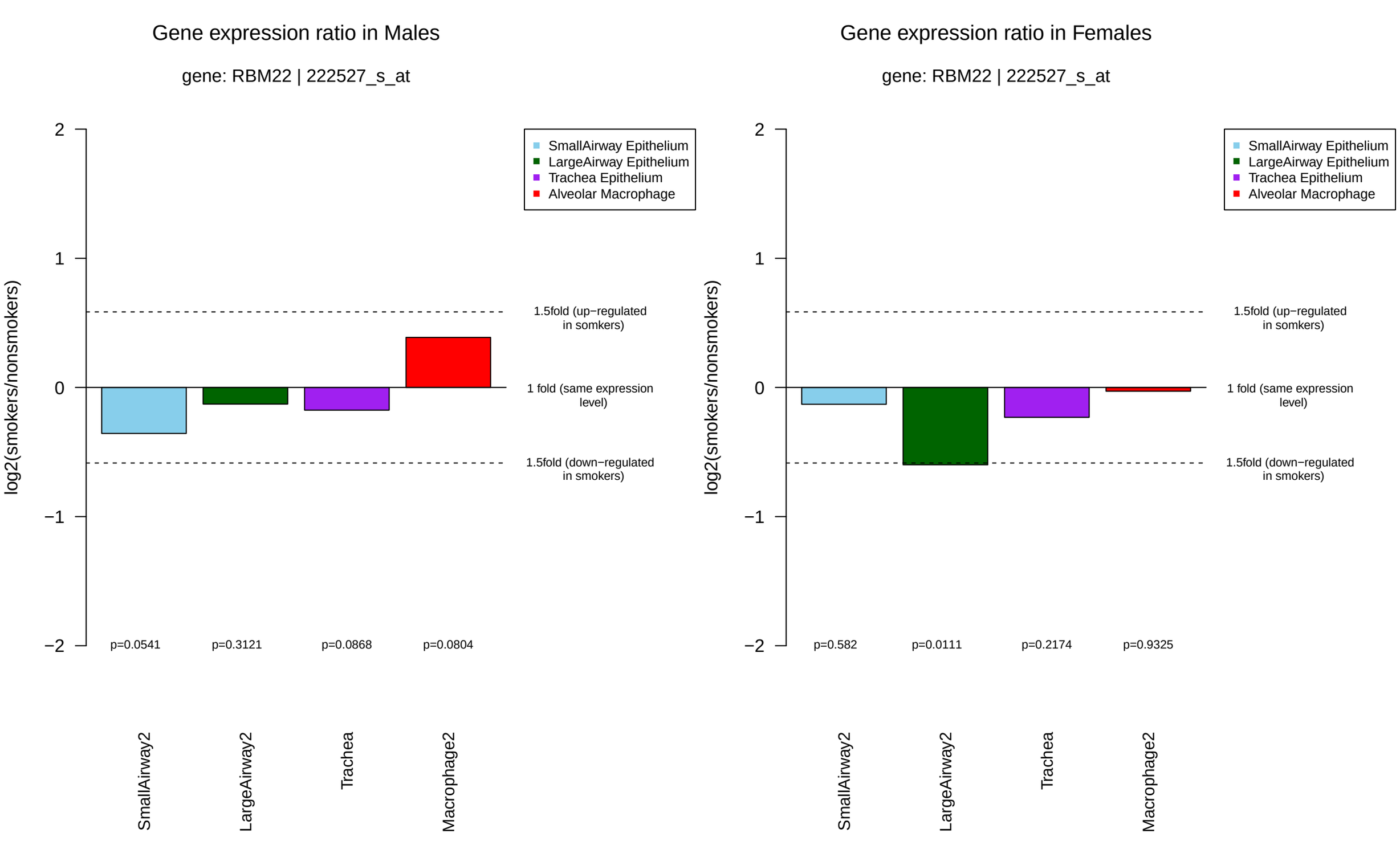
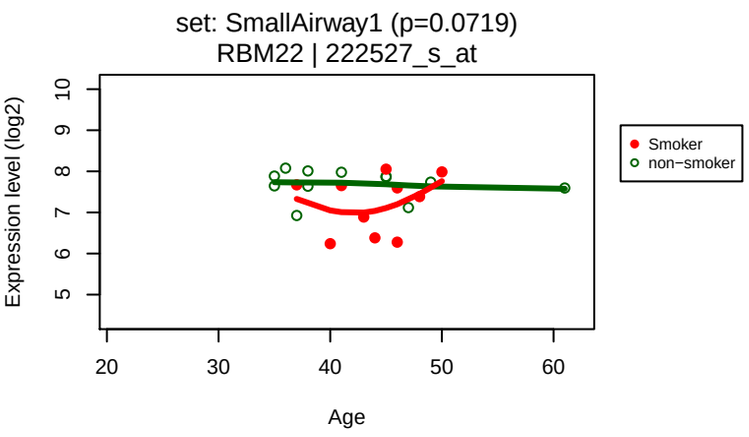
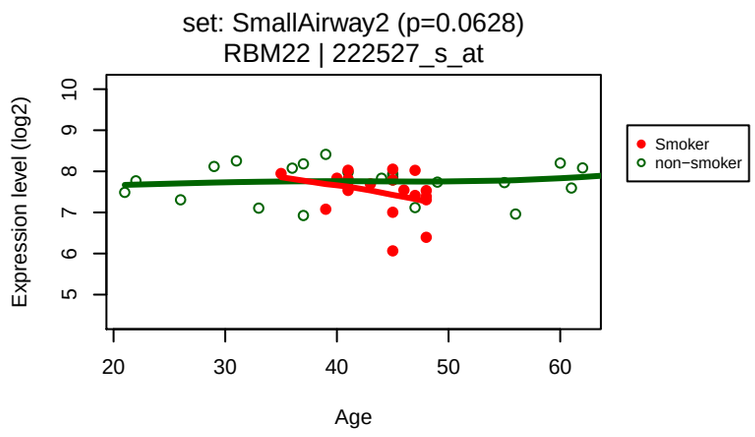
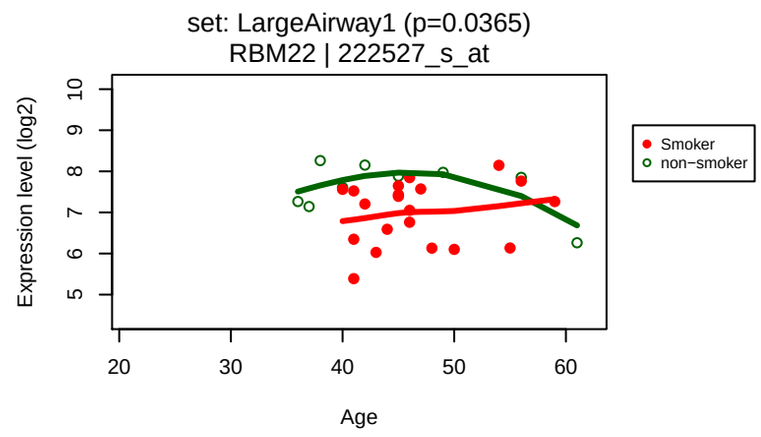

Gene expression of smokers and nonsmokers
gene: RBM22 | 222527_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0719
p=0.0628
p=0.044
p=0.0365
p=0.0087
p=0.0446
p=0.0987
p=0.1638
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

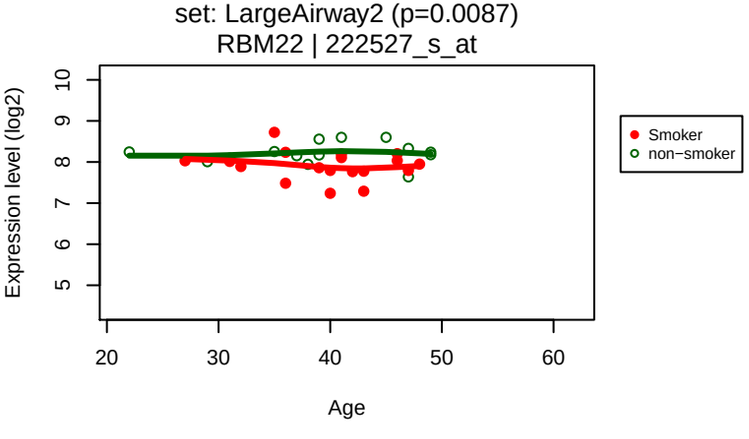

Gene expression ratio (smokers/nonsmokers)
gene: RBM22 | 222527_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0719
p=0.0628
p=0.044
p=0.0365
p=0.0087
p=0.0446
p=0.0987
p=0.1638
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

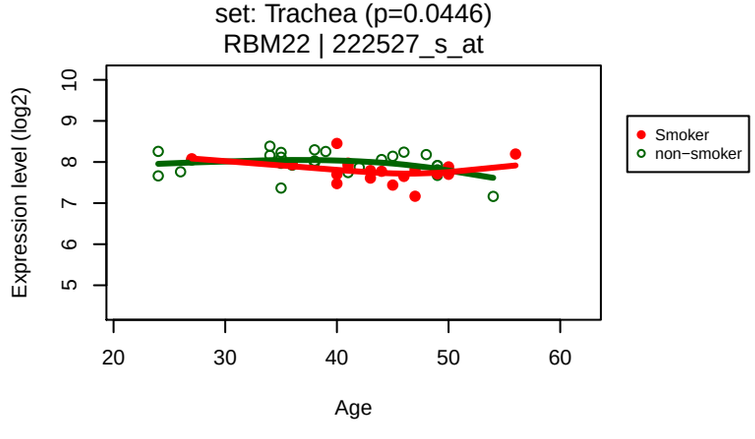

Gene expression ratio in Males
gene: RBM22 | 222527_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0541
p=0.3121
p=0.0868
p=0.0804
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RBM22 | 222527_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.582
p=0.0111
p=0.2174
p=0.9325
SmallAirway2
LargeAirway2
Trachea
Macrophage2

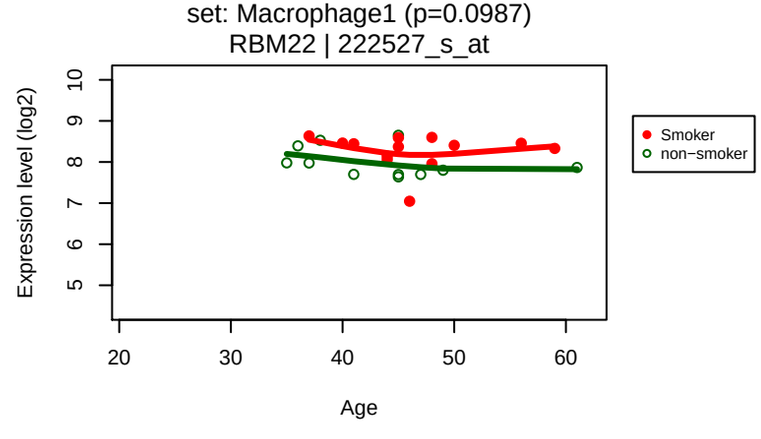

set: SmallAirway1 (p=0.0719)
RBM22 | 222527_s_at
Expression level (log2)
Age
Smoker
non−smoker

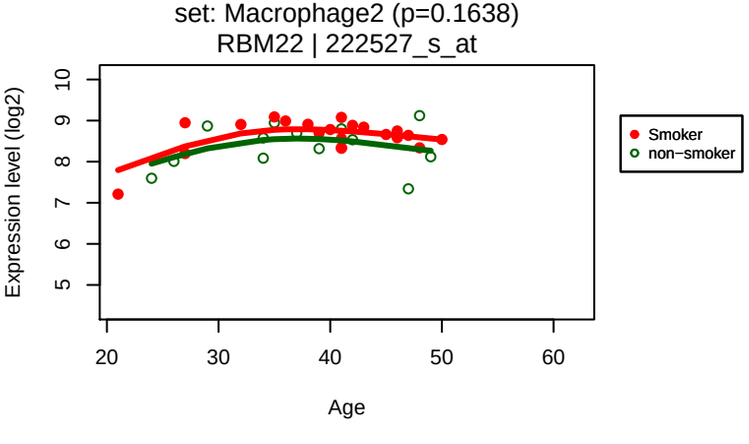

set: SmallAirway2 (p=0.0628)
RBM22 | 222527_s_at
Expression level (log2)
Age
Smoker
non−smoker

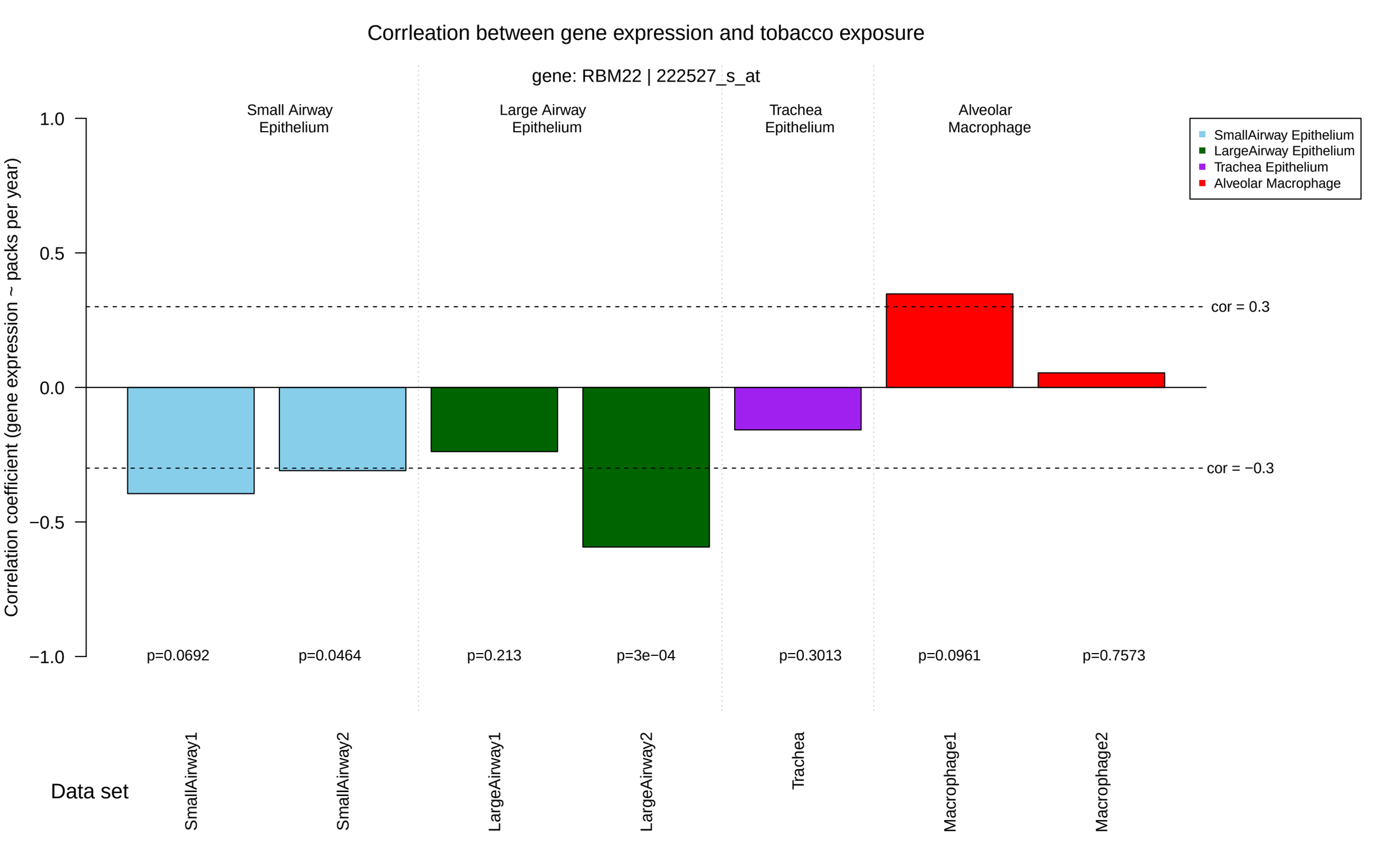

set: LargeAirway1 (p=0.0365)
RBM22 | 222527_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0087)
RBM22 | 222527_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0446)
RBM22 | 222527_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0987)
RBM22 | 222527_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1638)
RBM22 | 222527_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RBM22 | 222527_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0692
p=0.0464
p=0.213
p=3e−04
p=0.3013
p=0.0961
p=0.7573
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2